PROJECT – PART-I

Deliverable 1: Complete SCMP

Software Configuration Management Plan (SCMP)

Web-based Invoice Trading Platform

Prepared in accordance with IEEE Std 828-2012

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Document Control

Table 1: Document identification

Document field	Value
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Document Objective

This SCMP defines how software configuration items are identified, controlled, tracked and managed throughout the project lifecycle of the Web-based Invoice Trading Platform. It uses IEEE Std 828-2012 as its primary reference, addresses the plan-content areas in Annex D of that standard, and adapts them to this project. Approval and implementation remain subject to team verification; this document does not assert independently verified full conformance.

Revision History

Table 2: Revision history

Version	Change	Approval status
0.1	Initial complete SCMP draft.	Pending CCB review

Approval Record

Table 3: CCB approval record

Approval field	Actual record
CCB meeting / date	To be completed
Participating members and affirmative votes	To be completed
Decision and conditions	To be completed
Approved document version / commit	To be completed

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1. Introduction, Applicability and References

1.1 Purpose and Project

This plan defines how the five-member team identifies, acquires, stores, controls, tracks, audits, builds, releases and retires configuration items (CIs) for the Web-based Invoice Trading Platform. Its objective is a consistent, traceable, recoverable product configuration throughout requirements, development, testing, demonstration and final archival. It is written for the members who perform CM, the CCB that decides changes and the instructor who reviews the project.

The platform enables sellers to list unpaid invoices for discounted sale to investors. Planned capabilities include registration and role-based access, invoice upload and administration, browsing, bidding or purchase, payment/settlement status tracking and reporting. The approved SRS determines exact scope. Real fund movement, external payment integration and operational use are not assumed; introducing them requires approved requirements, design and change review.

1.2 Scope and Relationships

Control applies to frontend/backend code, database scripts, requirements/design/test/user documents, synthetic test data, build/deployment configurations, dependencies, interface contracts, release records and SCM evidence.

This SCMP defines governing processes. The companion Configuration Identification Document supplies the detailed naming and identification record; its inventory and rules must agree with this plan. The SRS, SDD, test plan and project schedule supply requirement, architecture, acceptance and milestone details when approved.

Affected teams and CI categories. Every member is affected in the frontend, backend, QA/documentation, build and SCM roles set out in Section 3.1. Controlled categories are documents (DOC), source (SRC), data (DAT), configuration (CFG), release (REL), records (REC) and third-party supplier items (SUP). The first six are listed in Section 4.2; supplier items are listed in CI-REC-03.

Relationship to the organization. This is a five-member academic project at Addis Ababa Science and Technology University, and no other project shares its CIs. The plan covers the product from requirements to final archive. Formality is deliberately light: unbaselined work is controlled by PR, and every baseline is controlled by the CCB. Deliverables at project close are listed in Section 14.1.

Level of this plan. This is a project-level plan. It gives actionable procedures for this team and repository; it does not set or repeat university or departmental policy. Section 2.4 records the outside rules it must respect.

1.3 Applicable References and Policies

Table 4: Applicable references and policies

Reference / directive	Application
IEEE Std 828-2012 [R1]	Primary process and plan-content reference; use the assigned 2012 edition.
NASA Software Engineering Handbook, topic 5.06 [R2]	Supplementary planning guidance only; NASA-specific directives are not imposed on this academic project.

Reference / directive	Application
Course Project Part-I instructions	Project-specific plan; identify CIs, naming/version rules, initial baseline and repository deliverables.
Approved project SRS, SDD and test plan	Scope, interfaces, required behavior and verification criteria; actual versions recorded in CI register.
Project security / contribution procedures	No secrets or real financial data in Git; independent reviews and controlled releases as defined below.
Software Configuration Management Handbook [R4]	Supplementary guidance on controlling and auditing the SCMP as a configuration item.
Personal Data Protection Proclamation No. 1321/2024 [R6]	Applies only if real personal data is proposed for the project; see Section 2.4.

1.4 Definitions

Table 5: Definitions and abbreviations

Term	Meaning
SCM / SCMP	Software Configuration Management / its governing plan
PM / QA	Project Manager / Quality Assurance (QA / Documentation Lead)
CM	Configuration Manager (role abbreviation used in estimate and resource tables)
CI	Configuration item: an artifact or explicit artifact grouping under control
CCB / CR	Change Control Board / Change Request
SRS / SDD	Software Requirements Specification / Software Design Document
Baseline	Reviewed and approved fixed configuration subject to formal change control
CI/CD	Continuous Integration / Continuous Delivery; distinct from CI meaning configuration item
FCA / PCA	Functional / Physical Configuration Audit
Interface contract	Controlled definition of information exchanged between components or external systems
PR	Pull request

1.5 Assignment Objective and Plan Map

The table below shows where this plan satisfies each element of the Project Part-I objective and notes.

Table 6: Assignment objective mapped to this plan

Objective element	Plan section	What the plan defines
CIs are identified	4	Selection criteria, CI inventory, storage, naming and version rules, baselines and control gates.
CIs are controlled	5, 8, 9	Branching and protection rules, Change Request procedure, CCB authority, interface control and supplier item control.
CIs are tracked	6, 7	Status accounting records, reports, management measures, functional and physical configuration audits.
CIs are managed throughout the lifecycle	2, 3, 10–13	Tailoring, organization, build and release, backup and recovery, retirement, schedule, resources, risks and plan maintenance.

Objective element	Plan section	What the plan defines
IEEE 828-2012 as primary reference	1.3, Appendix A	Applied reference and coverage matrix against the standard's clauses and Annex D.
Adapted to the actual project	1.1, 2, 4.2–4.4, 12	Platform-specific CIs, repository layout, technology, naming, roles, limits and estimates for this team.

1.6 Documentation Hierarchy and Required Records

SCM documentation is organized in five levels, with this SCMP as the master document.

Table 7: SCM documentation hierarchy

Level	Project documents	Reference location	Update responsibility
SCMP	ITP-SCMP-001 (CI-DOC-04): master document governing SCM activities.	docs/scm/	Project Manager; Configuration Manager as custodian
Policies and standards	IEEE Std 828-2012, course instructions and the obligations in Section 2.4.	Cited in Sections 1.3 and 2.4	Project Manager
Procedures and guidelines	Sections 4 to 11 of this plan, repository README and contribution rules (CI-CFG-05).	This plan; repository root; .github/	Configuration Manager
Supporting documents	Appendix B forms and checklists, issue and PR templates, training material and tool setup notes.	.github/; docs/scm/	Configuration Manager; Build Engineer
Records and reports	CI register, baseline manifests, CRs, CCB decisions, status, surveillance and audit reports, distribution records.	docs/scm/registers/, baselines/, ccb/, reports/; GitHub Issues; release/	Configuration Manager; QA for audit and surveillance reports

Revision practice. Documents follow the version rules in Section 4.4 and carry a revision history. The authoritative copy is the one at the applicable approved tag in the repository; copies shared through Telegram, email or print are uncontrolled.

2. Tailoring, Limitations and Assumptions

2.1 Tailoring Approach

This plan uses one CCB and one repository because the team has five members and shared roles. GitHub Issues, pull requests and controlled Markdown/Word records provide lightweight evidence. The tailoring register records the application decision and rationale; it must not be interpreted as silently deleting standard requirements. The CCB and instructor review any claimed non-applicability before approval.

Tailoring reduces ceremony, not control. Whatever the team size, configuration identification, change control, status accounting, auditing and independent review remain in force.

2.2 Tailoring Register

Table 8: Tailoring register

Area	Project application / limitation

Area	Project application / limitation
Lifecycle	Requirements through demonstration, maintenance during the course and final archive. Continued live operation requires an approved plan extension and named owner.
Hardware	Personal laptops and hosting are supporting resources; no custom hardware product is planned. Reproducible environment definitions remain controlled. No physical CIs are planned; any later physical item is labeled and stored under this plan.
Suppliers	No contracted developer is currently identified. Supplier controls apply to external libraries, tools and services; contracted-item controls become applicable if procurement occurs.
Operational data	Use synthetic data; no real debtor/customer financial records in repository or demonstration fixtures.
Payments	Track statuses as approved; no unapproved live payment provider or real transactions.
Tool capability	Verify branch protections and CI/environment gates in the selected GitHub setup. Use documented manual gates if unavailable.
Schedule / budget	Only assignment dates are known. Later milestones are phase-relative; effort and resource estimates below are proposed planning allowances.
Independence	Members hold multiple roles. No one is the sole reviewer of their own change or audit evidence.
AI/ML	Excluded from current product scope. Addition requires a CR and updated design/dependency controls.

2.3 Assumption Management

The Project Manager resolves missing course milestones and institutional policies before the relevant baseline. Assumptions are checked at every phase boundary; any invalid assumption is logged as a risk or CR.

Table 9: Planning assumptions

ID	Assumption	Response if it proves false
A1	The five members and the roles in Section 3.1 remain in place for the course.	Reassign roles under Section 3.3 and record the change before it takes effect.
A2	Each member has a GitHub account, a working laptop and enough internet access for weekly work.	Project Manager arranges shared access and adjusts assignments.
A3	The selected GitHub plan supports the branch rules in Section 5.2.	Use the documented manual gates in Section 5.2 and log them as deviations.
A4	Demonstrations use synthetic data and no live payments.	Raise a CR and complete the review in Section 2.4 before any real data or funds are used.
A5	Course milestones after November 12, 2026 are issued in time to schedule baselines.	Project Manager requests the dates and updates Section 12.2.
A6	Existing or free resources are sufficient for hosting and backup.	Treat as a resource gap under Section 12.3.

2.4 Policy and Regulatory Alignment

- **University and course:** instructor rules on submission format, dates and assessment take precedence over this plan. Any conflict is reported to the Project Manager and corrected through Section 13.1.

- **Personal data:** using synthetic data keeps real personal data out of the project. Before any real personal data is admitted, the Project Manager obtains a review against the Ethiopian Personal Data Protection Proclamation No. 1321/2024 [R6] and a CR is approved.
- **Financial activity:** the platform is not operated as a live invoice-trading service. Live operation requires a separate review of financial-sector and commercial requirements, including any National Bank of Ethiopia requirements, and a plan extension.
- **Licensing:** third-party items follow Section 9.3.
- **Documentation compliance:** records required by these rules are held as CIs under this plan, and no regulatory filing is claimed for this academic project.

3. Organization, Responsibilities and Authority

3.1 Team Roles and Authority

Table 10: Team members, roles and authority

Member	ID	Role and authority
Arsema Tefera	ETS0160/15	Project Manager and CCB Chair: owns plan, resources, schedule and decision records; authorizes emergency exception only as specified.
Arsema Tewodros	ETS0161/15	Configuration Manager: repository administration, CI register, baseline/release tags, reports, backup and configuration retrieval.
Arsema Tsegay	ETS0167/15	Backend Lead: backend/database reviews, backend dependencies and API contract co-ownership.
Azarya Melkamu	ETS0187/15	Frontend Lead / Build Engineer: frontend review, shared API contract co-ownership, automated builds and deployment.
Anasimos Derbe	ETS0151/15	QA / Documentation Lead: tests, traceability, audit coordination, independent verification and document control.

3.2 Reporting Chain and CCB Authority

- **Reporting chain:** technical leads and QA report SCM issues to the Configuration Manager, who reports status and exceptions to the Project Manager and CCB.
- **External advisor:** the instructor is an external advisor and receives baseline/release reports.
- **CCB membership:** the CCB includes all five members.
- **Approval rule:** approval requires at least three affirmative votes including either the Project Manager or Configuration Manager. The chair records but cannot unilaterally override the decision.
- **Decision record:** record attendees, votes, conditions, rationale and dates.
- **CCB purpose and objectives:** protect baseline integrity by evaluating and deciding proposed changes on technical and managerial grounds. All five members together represent every functional area affected.
- **CCB period and scope:** effective from approval of this plan until final handover. Its authority covers all baselines, scope-affecting changes, releases and this plan (Section 3.4).
- **CCB procedure:** the CCB meets at least every two weeks and sooner for urgent CRs. All members review the documented impact analysis, and votes may be recorded in the CR.

- **Problem resolution:** defects and SCM problems are assigned to the owning lead and verified by QA; unresolved disputes go to the CCB.

3.3 Backup and Tool Responsibilities

- All members contribute through reviewed branches.
- Configuration Manager and Project Manager back up each other's repository/record duties; frontend and backend leads back up build coordination; a non-author lead backs up QA verification.
- Any role reassignment is recorded before it takes effect.
- The Configuration Manager maintains the SCM tools; the Build Engineer maintains build tools; QA monitors process evidence.

3.4 Governance and Decision Boundaries

Each CI owner (Section 4.2) is accountable to the Configuration Manager for record accuracy and to the Project Manager for delivery. QA provides oversight of process compliance. Decision authority is limited as follows.

Table 11: Decision authority and boundaries

Decision	Authority	Boundary or escalation
Routine change to unbaselined work	Non-author reviewer through PR	Goes to the CCB if it affects scope, commitments or a shared interface.
Change to a baselined CI or agreed scope	CCB, using the Section 3.2 vote	CR required; no implementation before the decision is recorded.
Payment, settlement or authentication change	Two independent non-author reviewers	CCB decision is also required if the item is baselined.
Baseline or release approval	CCB	Configuration Manager applies the tag only after approval is recorded.
Emergency fix	Project Manager authorizes; Technical Lead implements	CCB review within two working days (Section 5.5).
Repository, access and SCM tool settings	Configuration Manager; Build Engineer for build tools	CR required if a baseline is affected.
Amendment to this SCMP	CCB	Follows Section 13.1.
Paid resources	Project Manager	Quote and approval recorded before use (Section 12.3).
Real funds, real data or live operation	CCB with instructor consultation	Outside current scope; requires a CR and Section 2.4 review.

3.5 Stakeholder Involvement

Table 12: Stakeholder involvement in SCM

Stakeholder	Type	Involvement	Channel
Project team	Internal	Author and review CIs, raise CRs, vote on the CCB.	GitHub, weekly meeting, Telegram
Course instructor	External	Advises, reviews tailoring, has read access and receives baseline and release reports. Does not vote on the CCB.	Repository access and course submissions

Stakeholder	Type	Involvement	Channel
Prospective sellers, investors and administrators	External	No live users during the course; their needs are represented by the SRS and acceptance tests.	Through QA and the Project Manager
Hosting, package and tool providers	External	Supply items and change notices handled under Section 9.	Provider notices; dependency advisories
Regulators	External	No active involvement while the Section 2.4 limits hold.	Project Manager, if live use is proposed

3.6 Communication and Reporting Structure

Table 13: SCM communication paths

Information	From → to	When	Channel
Daily coordination	All members	As needed	Telegram
Progress and blockers	Project Manager → team	Weekly	Team meeting
Configuration status report	Configuration Manager → team	Every two weeks	docs/scm/reports/
Surveillance findings	QA → Configuration Manager and Project Manager	Every two weeks	Status report section
CCB agenda	Configuration Manager → CCB	One working day before review	GitHub Issue
Exceptions and deviations	Any member → Configuration Manager → Project Manager and CCB	On discovery	GitHub Issue
Baseline and release reports	Configuration Manager → Project Manager and instructor	Each approval or release	Repository and course submission

A decision made in Telegram or at a meeting takes effect only once it is recorded in the linked GitHub Issue or CCB decision record.

4. Configuration Identification and Acquisition

4.1 Selection Criteria and Granularity

An item is selected as a CI when it:

- defines agreed scope or interfaces;
- is needed to build, test, deploy or recover the product;
- requires independent approval or ownership;
- is delivered to the instructor or users; or
- provides traceability and configuration evidence.

Group related source files by component while identifying exact files with Git paths and commits. Documents, interface contracts, migrations and release manifests are explicitly identifiable. Split a CI when ownership, approval, independent release or risk requires separate control.

CI structure. The hierarchy below follows the product architecture. The product contains no firmware or custom hardware.

Table 14: CI structure and hierarchy

Level	CIs	Relationship
Product	Each baseline and release	Composite CI: a versioned collection of the items below.
Application	CI-SRC-01, CI-SRC-02, CI-SRC-03	Frontend calls the backend through CI-DOC-12; backend persists data through the database schema.
Verification	CI-SRC-04, CI-DAT-01	Automated tests exercise the application using synthetic data.
Build and environment	CI-CFG-01 to CI-CFG-06	Produce builds and record the environment and tool chain used.
Documentation	CI-DOC-01 to CI-DOC-12	Describe the functional and physical characteristics of the product.
Third-party items	CI-SUP items in CI-REC-03	Incorporated into the application and build after acceptance (Section 9).
Release and records	CI-REL-01, CI-REC-01 to CI-REC-04	Record versions, decisions and evidence.

4.2 Inventory and Capture

Table 15: Configuration item inventory

CI ID	Item	Controlled location	Owner
CI-DOC-01	Charter	docs/project/	Project Manager
CI-DOC-02	SRS	docs/requirements/	QA
CI-DOC-03	SDD	docs/design/	Leads
CI-DOC-04	SCMP	docs/scm/	Project Manager
CI-DOC-05	Test plan/manual cases	docs/testing/	QA
CI-DOC-06	User manual	docs/user/	QA
CI-DOC-07	Configuration Identification Document	docs/scm/	Configuration Manager
CI-DOC-08	Traceability matrix	docs/testing/	QA
CI-DOC-09	UI design	docs/design/ui/	Frontend Lead
CI-DOC-10	Database design/dictionary	docs/design/database/	Backend Lead
CI-DOC-11	Test reports	docs/testing/reports/	QA
CI-DOC-12	Interface contracts	docs/design/interfaces/	Both leads
CI-SRC-01	Frontend	frontend/	Frontend Lead
CI-SRC-02	Backend	backend/	Backend Lead
CI-SRC-03	Schema/migrations/seeds	database/	Backend Lead
CI-SRC-04	Automated tests	tests/	QA
CI-DAT-01	Synthetic data	tests/data/	QA
CI-CFG-01	CI workflows	.github/workflows/	Build Engineer
CI-CFG-02	Environment templates	frontend/.env.example; backend/.env.example	Build Engineer

CI ID	Item	Controlled location	Owner
CI-CFG-03	Dependency manifests/lock files	frontend/; backend/	Leads
CI-CFG-04	Build/deploy scripts	scripts/	Build Engineer
CI-CFG-05	Repository governance	.github/; .gitignore	Configuration Manager
CI-CFG-06	Environment/tool version record	docs/scm/environments/	Build Engineer
CI-REL-01	Release notes/delivery manifests	release/	Configuration Manager
CI-REC-01	CRs/decisions	GitHub Issues; docs/scm/ccb/	Configuration Manager
CI-REC-02	CI register/baselines	docs/scm/registers/; docs/scm/baselines/	Configuration Manager
CI-REC-03	Supplier/dependency acceptance register	docs/scm/suppliers/	Leads
CI-REC-04	Status, surveillance and audit reports	docs/scm/reports/	Configuration Manager

New CI acquisition follows these steps:

1. The producer submits the item with its description, maintenance information and provenance.
2. The Configuration Manager allocates a unique ID and records the actual path, version and status.
3. The designated reviewer checks completeness, security and acceptance criteria.
4. Approved contents enter the repository through a PR; the first controlled commit and relevant baseline are registered.

Rejected items remain outside approved configurations with the rejection reason recorded. External items are recorded using immutable identifiers and controlled snapshots where licensing permits.

Register fields: ID, description, type, owner/backup, origin (date and source), path, approved versions, Git reference, status, baseline membership, parent and successor CIs, license where relevant, affected interfaces, related CRs with their implementation status and result, and last update.

Status values: Planned, Draft, Under review, Approved, Baselined, Released, Retired. Planned entries do not imply the artifact exists.

4.3 Planned Technology and Storage

Table 16: Planned technology stack

Area	Planned technology
Frontend	React.js
Backend	Node.js / Express REST API
Database	PostgreSQL
Authentication	Email/password with JWT-based authentication and role-based authorization; the SDD defines authentication/session details.
Version control	Git / GitHub
Automation	GitHub Actions
Testing	Jest / Supertest

Area	Planned technology
Documentation	Markdown and Word/PDF

Record actual runtime, database, package-manager and tooling versions in CI-CFG-06 before baseline use. Manifests and lock files capture dependencies. Any baseline-affecting tool change requires a CR.

Repository: the repository name is invoice-trading-platform. Main locations are docs/, frontend/, backend/, database/, tests/, scripts/, release/ and .github/. The README records setup, team, structure and current baseline.

Retrieval: exact configurations are retrieved through approved tags and manifests. Large external assets are stored with immutable version IDs, access details and approved snapshots/checksums where applicable; links alone are insufficient. Retrieval includes checking the tag, manifest and checksum. The Configuration Manager tracks controlled copies issued outside the repository, and security-sensitive items remain under restricted access.

Table 17: Controlled repositories

Repository	Format and location	Receiving and inspection	Access control
Source and documents	Git repository with the layout above; branching in Section 5.1.	PR, required checks and non-author review.	Section 5.6
Change records	GitHub Issues and PRs; exports in docs/scm/ccb/.	Configuration Manager checks required fields.	Team write; instructor read
Release packages	GitHub Releases or recorded controlled storage, with checksums.	PCA and CCB approval.	Configuration Manager publishes; team and instructor read
Backup archives	Git bundles, database dumps and exports in separate storage.	Restoration check at each baseline.	Configuration Manager and Project Manager only

Every item held in these repositories must have an entry in the CI register or, for third-party items, in CI-REC-03.

4.4 Naming and Version Identification

Table 18: Naming and version rules

Item	Rule / example
CI ID	CI-<category>-<number>; permanent identifier, e.g. CI-DOC-02.
Document	ITP-<Type>-<Title>-v<Major>.<Minor>.<extension>; ITP-SRS-Requirements-v1.0.docx.
Folders / general JS	Lowercase hyphenated: invoice-management/; invoice-service.js.
React components	PascalCase: InvoiceUploadForm.jsx.
Functions / constants	camelCase / UPPER_SNAKE_CASE.
Tests / migrations	invoice-service.test.js; 001_create_users.sql.
Branches	feature/invoice-upload; bugfix/login-validation; hotfix/payment-rounding; docs/update-srs; release/1.0.0.
Commits	<type>(<scope>): <description>; feat(invoice): add upload form. Types: feat, fix, docs, test, refactor, style, chore.
Records	FR-01; NFR-01; TC-01; CR-001; map CR to actual GitHub Issue.

Item	Rule / example
Software	MAJOR.MINOR.PATCH: incompatible interface/behavior changes; backward-compatible functionality; backward-compatible fixes. Pre-release: 1.0.0-rc.1.
Builds	<version>+<workflow run>.<short commit>; 1.0.0-rc.1+42.a1b2c3d.
Third-party items	CI-SUP-<number> with supplier name and exact version retained; CI-SUP-001 for express@<locked version>.
Documents	0.x draft; 1.0 first approval; 1.1 minor approved update; 2.0 substantial approved revision.
Exact source	CI grouping, path, commit and applicable baseline/release tag. No independent version required for every file.

- Editable documents are authoritative; exported PDFs must match the approved revision.
- Applied baselined migrations are corrected with new migrations.
- Never move or reuse an approved baseline or release tag. New approved configurations receive new revision/release identifiers.

4.5 Baselines and Control Gates

Each baseline is an approved, fixed reference point. It records what the team agreed at the end of a phase, gives later changes something to be measured against, and allows that configuration to be rebuilt. The CCB approves it, the Configuration Manager records it and QA confirms its evidence. Documents and code are linked through the baseline manifest (Appendix B.2).

Table 19: Baselines and control gates

Baseline	Git tag	Entry contents and approval gate
BL-00	ITP-BL-00-initial	Actual reviewed starting charter, SCMP, identification document, CI register, README, governance and available starter code/configuration; CCB approval.
BL-1	ITP-BL-01-requirements	Reviewed charter/SRS and requirement identifiers; CCB approval.
BL-2	ITP-BL-02-design	Reviewed SDD, UI/database design and interface contracts; CCB approval.
BL-3	ITP-BL-03-development	Implemented scoped code, applicable passing unit/API tests and updated test plan; phase audit and CCB approval.
BL-4	ITP-BL-04-testing	System/acceptance evidence, traceability and FCA findings resolved/dispositioned; CCB approval.
BL-5	ITP-BL-05-release	Final source/docs, release notes, delivery manifest and PCA; CCB approval. Software tag such as v1.0.0 also recorded.

All repository submissions receive PR control from capture; formal baseline changes require CR control after their gate. The Configuration Manager prepares each manifest with actual CI IDs, versions/paths, exact commit, approvals, checks and tag. After approval, tag and back up the fixed configuration. Baseline tags above are proposed until verified. Include only existing reviewed artifacts; empty directories or planned items are not completed deliverables.

4.6 Traceability Structure

Each requirement is traceable along this chain: requirement (FR/NFR) → design element or interface contract → source path and commit → test case (TC) → test result → related CR → baseline or release.

Table 20: Traceability records

Link	Recorded in	Maintained by	Checked at
Requirement → design	Traceability matrix (CI-DOC-08)	QA with leads	BL-2
Requirement → code	PR reference to FR/NFR ID; CI-DOC-08	Technical leads	BL-3
Requirement → test and result	CI-DOC-08; test reports (CI-DOC-11)	QA	BL-3 and BL-4 (FCA)
CR → affected CIs → PR and commit	CR records (CI-REC-01)	Configuration Manager	Closure of each CR
CI → baseline or release	CI register and manifests (CI-REC-02)	Configuration Manager	Each baseline; PCA at BL-5

Before BL-4, every baselined requirement must have at least one design link and one test case, or an approved deferral. A missing link is recorded as an audit finding. Appendix C shows the record format.

5. Configuration Change Control

5.1 Branching Model

The project uses simplified GitFlow:

- feature/docs branches start from develop; reviewed work returns to develop.
- release/* starts from develop and accepts fixes/documentation only. Approved releases merge to main; release fixes return to develop.
- Urgent hotfix/* starts from main and merges to both main and develop.

5.2 Branch Protection

- Protect main/develop against direct pushes, force pushes and deletion.
- Require passing checks and a non-author approval.
- Payment, settlement and authentication changes require two independent non-author approvals.
- Configure enforceable controls and record any manual gate needed by the selected hosting plan; administrative bypass must be logged and reviewed.

5.3 Change Request Procedure

1. Submit an issue/CR with affected CIs, baseline, description, justification, requester, priority and target date.
2. Configuration Manager and relevant leads assess functional, interface, security, test, document, effort, schedule and dependency impacts.
3. CCB approves, rejects or defers baselined/scope-affecting changes using the Section 3.2 voting rule; record rationale and conditions.
4. Implement approved work on a linked branch with focused commits and a PR.
5. Run required checks; independent reviewers examine code, interface compatibility, documentation and evidence.

6. QA verifies the accepted change; Configuration Manager updates records and any approved baseline revision.
7. Close only after acceptance evidence, related updates and configuration status are recorded.

Evaluation criteria. The CCB evaluates each CR against: need and fit with baselined scope; severity of the problem and how widely it affects users; effect on interfaces, security, data and tests; effort and schedule against remaining capacity; and whether the change can be verified and reversed.

5.4 Change Request Workflow

CR states: Open → Evaluating → Approved/Rejected/Deferred → In progress → Verified → Closed.

Routine unbaselined changes use issue/PR review, unless they affect agreed scope or commitments. A rejection or deferral retains its rationale. Implementation commits, PR, tests and final configuration are linked to the CR.

5.5 Emergency Changes

A Technical Lead (Backend or Frontend) may implement an urgent security/release fix with recorded Project Manager authorization; preserve tests and independent review where practicable, identify affected users/configurations, and obtain CCB review within two working days. Rejected emergency fixes require controlled remediation/rollback, never silent history rewriting.

5.6 Access and Issue Tracking

- **Access:** Project Manager and Configuration Manager administer the repository; developers/QA have write access subject to branch controls; the instructor has authorized read access.
- **Accounts:** use personal accounts and two-factor authentication. Review access at milestones and membership changes.
- **Issue labels:** bug, feature, documentation, change-request, security and priority labels; defect severity is tracked separately.
- **Board:** Backlog, In Progress, In Review, Testing, Done.

5.7 Change Records and Disposition

Table 21: Required change attributes

Attribute	Constituent CI (commit and PR)	Baseline (CR and CCB)
Originator / attribution	Commit author and issue author.	Requester and the accountable CCB.
Impact analysis	Git merge-conflict check and PR review.	Documented and reviewed by all CCB members, including severity, reach and project impact.
Rationale and approval	Commit or PR cites the CR or Issue number.	CCB decision with reasons.
Notification	CI owner requested as reviewer; GitHub notifies subscribers.	Affected parties notified through the CR and status report.
Reversibility	Git revert.	Earlier approved tag, with constituent changes individually reversible.
Set of changes	Commit and PR file list.	Affected baseline and impacted CIs listed in the CR.

Attribute	Constituent CI (commit and PR)	Baseline (CR and CCB)
Audit trail	Git history retained; no history rewriting on protected branches.	CR forms retained and exported at each baseline.

- **Unsuccessful changes** are backed out or corrected before a new baseline is established.
- **Rejected or deferred CRs:** the Configuration Manager records the disposition and verifies that the change is absent from the controlled configuration.

6. Configuration Status Accounting and Management Monitoring

6.1 Status Records

The Configuration Manager maintains the CI register, baseline/release history, CR log, approvals, defect status and build/test results from Git, Issues and PR records. All members can read status evidence; sensitive incident details use restricted access. Preserve links and dated exports at baseline and final archive; correct records by traceable amendment, not undocumented replacement.

Information needs. At the start of each phase the Configuration Manager confirms with the Project Manager and team what status information is needed and how often, including on-demand reports. Records must at least show each CI status, the current baseline, the state of every CR and two-way traceability (Section 4.6). The Configuration Manager also verifies that the tools capture these data and can be searched for reports.

6.2 Status Reports

Table 22: Status reports

Record / report	Timing	Recipient
Configuration status report	Every two weeks	All team members.
CCB decision record	After each decision	Team.
Baseline manifest/report	Each approval	Project Manager and instructor.
Release report	Each release	Project Manager and instructor.
Exception/corrective action	On discovery; tracked to verified closure	Responsible owner and CCB.

6.3 Management Measures

Table 23: Management measures and triggers

Measure	Definition / proposed management trigger
Baseline completeness	Included CIs with verified version/path/approval ÷ all baseline CIs; require 100% before approval.
Required-check pass rate	Passing required CI runs ÷ completed required runs over reporting period; show counts and cancelled runs separately. Any failed required run blocks its merge.
CR age	Working days since submission; pending >5 working days escalates to chair for decision scheduling.
Audit findings	Open count by severity and age; no unresolved critical/high release defect at delivery.
Approval compliance	Merged PRs meeting review rules ÷ merged PRs in reporting period; target 100%; investigate each exception.

Measure	Definition / proposed management trigger
Recovery verification	Date, configuration restored, result and elapsed restoration time; failed verification blocks release backup acceptance.

These targets are the performance goals of this plan. Service levels are: CCB decision scheduled within five working days of submission, a status report every two weeks, and a backup and restoration check at every baseline.

6.4 Management Monitoring

- QA samples evidence at milestones and through surveillance (Section 6.5), and reports deviations.
- The Project Manager compares actual effort, delays and resources against Section 12 estimates every two weeks.
- The CCB assigns corrective actions with owner, due date and verification evidence.
- Targets and triggers here are project planning rules, not quoted IEEE thresholds.
- Resource shortfalls and resulting risks are reported to the Project Manager. The Project Manager and Configuration Manager decide on any residual risk, and this plan is updated when mitigation is agreed.
- Third-party deliveries are tracked with the schedule, and the risk register is reviewed at the same two-week checkpoint.

6.5 Surveillance and Oversight

Surveillance is routine monitoring of whether this plan is being followed, so that deviations are found early. It differs from the formal audits in Section 7, which check a specific baseline at a planned gate. QA leads surveillance. The Configuration Manager samples any evidence that QA authored.

Table 24: Surveillance activities

Activity	What is checked	Frequency	Reported to
Repository and baseline review	Tags unchanged; register matches repository; no direct pushes to protected branches.	Every two weeks	Configuration Manager
CR log monitoring	Open CR age, missing impact analysis or decision, closure evidence.	Every two weeks	Project Manager
Workflow adherence	Sample of five merged PRs, or all if fewer: linked issue, non-author approval, passing checks.	Every two weeks	Configuration Manager
Security and dependency hygiene	No secrets or real financial data in commits; lock files updated.	Each iteration	Technical leads
Recurring issue review	Repeated findings and their causes.	Each phase boundary	CCB

Results are included in the status report (Section 6.2). Deviations are handled under Section 13.2, and recurring issues receive a CCB corrective action.

7. Configuration Audits and Reviews

7.1 Audit Planning

QA schedules milestone audits before BL-3, BL-4 and BL-5. Owners provide artifacts; non-author reviewers verify evidence. Audit plans identify scope, configuration, reviewers, criteria and date. Findings record severity, affected CI, evidence, corrective action, owner and closure verification.

7.2 Functional Configuration Audit (FCA)

The FCA traces baselined requirements to design, code and test cases, traces test cases to their execution results, and reviews functional and non-functional acceptance criteria, results and approved deferrals.

- BL-3 audits cover implemented scope only and do not claim final acceptance of unfinished work.
- BL-4 reviews full system/acceptance evidence.
- Release confirms no regression affecting acceptance.

7.3 Physical Configuration Audit (PCA)

The PCA compares actual source, documents, dependencies, packages and interface records with the manifest and tag. It verifies names/versions, build attributes, change records (rationale, impact analysis, approval, reversibility and attribution), release notes (known defects, test limitations and version identification), excluded secrets, delivery contents and clean-environment rebuild, and confirms that externally stored artifacts are retrievable and identified. The release PCA is performed before BL-5.

7.4 Audit Reporting and Closure

QA issues the audit report to the Project Manager, CCB and instructor; leads resolve findings, an independent reviewer re-audits the affected area and reports the result, and the CCB accepts the disposition before baseline approval. Release requires no unresolved critical/high defects. Record any accepted lower-severity deviation and its rationale in release notes and approval records.

7.5 Baseline Configuration Audit

A baseline configuration audit is performed for every baseline from BL-00 to BL-5, before the CCB decision. A non-author reviewer confirms that only CIs in the required state are included, that the baseline and its CIs are correctly identified in the manifest, and that the required approvals are recorded. Results follow Section 7.4.

7.6 Integration with Quality Objectives

SCM supports the quality objectives below. Test strategy and detailed quality procedures remain in the test plan and are not repeated here.

Table 25: Quality objectives supported by SCM

Quality objective	SCM support	Measure (Section 6.3)
Product meets approved requirements	Requirement baselines, traceability and FCA.	Baseline completeness; audit findings
Product can be rebuilt and recovered	Tagged baselines, lock files, PCA rebuild and restoration tests.	Recovery verification; required-check pass rate

Quality objective	SCM support	Measure (Section 6.3)
Changes are reviewed	CR control, branch rules and independent review.	Approval compliance; CR age
Documents match the product	Same-change document updates and controlled revisions.	Audit findings on documents
Financial and credential data is protected	Synthetic data, secret handling and two reviewers for payment, settlement and authentication changes.	Approval compliance; audit findings

8. Interface Control

8.1 Ownership and Contract Content

Both technical leads own shared frontend/backend API contracts; the Backend Lead owns database interfaces. Any external provider boundary also has a named project owner. Controlled contracts include endpoints/methods, request/response schemas, authentication/authorization, errors, data types, identifiers, financial amount/currency representation, rounding, date/time handling and compatibility rules as applicable.

Table 26: Key product interfaces

ID	Interface	Nature	Affected parties	Specification
IF-01	Frontend ↔ backend REST API	Software / data	Frontend and Backend Leads	CI-DOC-12
IF-02	Backend ↔ PostgreSQL database	Data	Backend Lead	CI-DOC-10; CI-SRC-03
IF-03	Users ↔ web interface	Human-system	Frontend Lead; QA	CI-DOC-09
IF-04	Repository ↔ automation and hosting	Software	Build Engineer; service provider	CI-CFG-01; CI-CFG-02

No separate interface CCB is established; the project CCB controls all interface baselines.

8.2 Interface Change Procedure

1. Identify each interface and participating components/providers in CI-DOC-12.
2. Review and approve initial contracts with design baseline BL-2; record contract version and consuming component versions.
3. For proposed changes, assess all producers/consumers, migrations, security and tests; use CR control after baseline.
4. Obtain both internal lead reviews for shared API changes and coordinate any external party before implementation.
5. Implement producer/consumer updates and contract/integration tests together or record a compatible transition plan.
6. Verify the approved contract against implementation and update traceability, test evidence and release compatibility notes.

8.3 Incompatible Changes and Escalation

Incompatible changes require explicit version/transition decisions and CCB approval; do not silently break an approved client or data interface. Provider change notices are evaluated through the same impact process. Interface disagreements are escalated to the CCB before merge.

9. Supplier Configuration Item Control

9.1 Supplier Scope

External libraries, runtimes, tools, hosted services and externally supplied artifacts are evaluated as supplier items. No contracted development supplier is currently identified; adding one requires recorded acceptance/delivery responsibilities and change-notification arrangements before acquisition. Library and service providers do not take part in project change management; their release notes and notices are assessed through Section 5.

9.2 Supplier Controls

Table 27: Supplier item controls

Control	Project procedure
Selection	Lead checks purpose, provenance, maintenance, supported version, compatibility, vulnerabilities, license and redistribution obligations.
Identification	Register supplier/package, version or immutable ID, origin, license, affected CIs, owner, acceptance evidence and update method.
Acceptance	Review manifests/lock files, run relevant tests, inspect required notices and reject unapproved or incompatible items before baseline inclusion.
Updates	Use reviewed PR; baseline-affecting/security/interface changes receive CR review. Record old/new versions, reasons and impact.
Services	Record configuration/API version where available, account owner, dependency and backup/export limitations; do not store credentials in Git.
Monitoring	Check advisories/dependencies each iteration and before release; triage findings with owner and disposition.
Replacement/withdrawal	Assess replacement, migration, rebuild and test impacts; obtain approval and retain prior identification evidence.

9.3 Licensing and Contracted Suppliers

- License review must not assume that academic or open-source use permits every redistribution. Keep required notices.
- If a contract supplier is added, specify CM requirements in the agreement (including a supplier CMP), delivered item identifiers, supplier baseline information, acceptance tests, compliance monitoring, supplier audits, evidence access, handling of proprietary items and responsibility for supplier changes.
- Supplier failures become tracked defects/risks, not unrecorded substitutions.

10. Build, Release and Delivery Management

10.1 Reproducible Builds

The Build Engineer controls workflow scripts, runtime/database/package-manager version records, lock files and environment templates. GitHub Actions runs applicable dependency installation, lint, unit/API tests and production build for pushes/PRs. Record source commit, tool versions, workflow run, test evidence and artifact identity. Failures block required merges. Secrets stay in approved secret storage or ignored local files; templates contain placeholders.

Table 28: Build environments

Environment	Use	Source
Local	Individual work	Task branch.
Staging	Integration/system tests	The develop branch after passing checks.
Production/Demo	Approved demonstration delivery	Approved tagged main commit only.

10.2 Release Authorization and Distribution

1. Create release/x.y.z from develop with scope and candidate commit.
2. Allow fixes/documentation only; run regression/acceptance tests and migrations including applicable upgrades.
3. Complete PCA, defect review, traceability and document updates.
4. CCB approves the identified candidate and delivery conditions. Any material subsequent change requires renewed approval.
5. Merge the approved release to main; verify final contents/checks and apply an immutable release tag; record final commit and manifest.
6. Use a release gate to verify approval before deploying that tagged commit. Merge to main alone does not authorize deployment.
7. Configuration Manager publishes the identified package/release notes through the controlled delivery location and records recipients/access.
8. Merge release fixes back to develop; update status, baseline and backup records.

10.3 Release Checklist

- Requirements implemented or formally deferred.
- No critical/high open defects; required tests pass.
- Migrations verified.
- User/setup documents current; licenses/notices present.
- Known issues disclosed.
- CCB decision recorded.
- Tag, manifest, package identity and backup verified.

Store notes in release/ and packages in GitHub Releases or recorded controlled storage. Include checksums for distributed packages and version/commit identifiers in manifests.

10.4 Delivery Acceptance and Rollback

Delivery acceptance records recipient/instructor acknowledgement or demonstration result, delivered version, date and outstanding conditions. Failed acceptance returns to tracked remediation.

Rollback identifies the previous approved application configuration and a database-compatible recovery strategy; reverting application code does not automatically reverse migrations.

10.5 Release Types, Format and Tracking

Table 29: Release types

Release type	Contents	Recipient and trigger	Format and distribution
Internal test	Build from develop.	QA, after checks pass	Staging deployment
Release candidate	Build from release/x.y.z and draft release notes.	QA and team, after scope freeze	Tagged pre-release in GitHub Releases
Final demonstration	Approved build, source, release notes, manifest, and setup and user documents.	Instructor, after CCB approval	Compressed package with checksum in GitHub Releases; deployment from the tagged main commit

Each approved release is identified and baselined as a CI. CI-REL-01 records its date, recipients, state (Candidate, Approved, Delivered, Superseded or Archived), environment (operating system, runtime and database versions) and the release it supersedes. These records are available to the team and instructor.

11. Preservation, Recovery, Security and Retirement

11.1 Repository Backup

GitHub is the master working repository. The Configuration Manager creates a Git bundle with all relevant refs monthly and at every baseline, stored separately. Back up release assets, issue/decision exports, external artifacts and any large-file objects separately because a Git bundle alone does not capture them. Restrict backup access; record backup date, scope, location and verification. Label each backup with the project ID, baseline or release identifier and date, including any physical media.

11.2 Database Backup and Retention

While hosted environments are active, create daily database dumps with synthetic data and test restoration before each release. Proposed retention is the latest 30 daily database backups and all baseline archives through final submission plus the course review period; confirm its actual end date with the instructor before deleting anything. Preserve credentials through a separate authorized secure mechanism, never in repository backups.

11.3 Recovery Procedure

1. Log the incident and identify the last approved configuration.
2. Restore Git and external artifacts, then restore a compatible database.
3. Rebuild, test and verify integrity.
4. Obtain authorization before shared redeployment.
5. Record recovered version, elapsed time, lost changes and corrective actions.

Exposed secrets must be revoked/rotated immediately, impact assessed and evidence recorded; deleting history alone is insufficient.

11.4 Final Handover and Retirement

At final handover, archive approved source/docs/manifests, release packages, records and restoration instructions. Transfer ownership if maintenance continues. If retiring the demonstration service, obtain authorization, export required records, revoke service credentials, remove hosted synthetic data as appropriate and record closure. Preserve the approved archive and master copies for the agreed retention period. When a release reaches end-of-life, archive it in the designated location, remove it from normal distribution and mark it Archived in its release record.

12. Activities, Dependencies, Schedule and Resources

12.1 Activity Estimates

The following values are proposed team planning allowances, not historical measurements or IEEE-prescribed effort. One person-hour is one person working for one hour; elapsed duration depends on availability. The team approves the number of iterations, milestones and CRs as the course schedule becomes known.

Table 30: SCM activities, dependencies and estimates

Activity	Owner	Predecessor → output	Effort / elapsed allowance
Plan/inventory review	PM + CM	Project scope available → reviewed plan/inventory	10 person-hours / 2 working days
Repository/control setup	CM + Build Engineer	Plan draft → verified repository/PR controls	6 person-hours / 1 working day
Initial baseline	CM + QA + CCB	Reviewed actual artifacts/setup → BL-00	5 person-hours / 1 working day
Requirements/design baseline	Leads + QA + CM	Relevant reviewed phase outputs → BL-1 or BL-2	4 person-hours per baseline / 1 working day each
Status accounting	CM	Current records → fortnightly report	1 person-hour per two-week period
CCB review	All members	Impact analysis → decision record	2.5 person-hours per meeting (5 × 0.5 hour)
Dependency check	Leads	Dependency register → findings/decision	1 person-hour per iteration
Milestone audit	QA + reviewer	Test/build evidence → findings/closure	4 person-hours per BL-3, BL-4 or BL-5 audit / 1–2 working days
Baseline audit	QA + non-author reviewer	Manifest and included CIs → audit record	1 person-hour per baseline, within the baseline allowance
Release coordination	CM + Build Engineer	Tests/audit/CCB approval → delivery record	6 person-hours per release / 1–2 working days; audit effort separate
Baseline backup verification	CM	Approved baseline → verified archive	1 person-hour per baseline

Activity	Owner	Predecessor → output	Effort / elapsed allowance
Training	CM + team	Repository setup → attendance/practice record	5 person-hours total / 1 hour elapsed
Final archive	CM + QA	Accepted release → handover/retirement record	3 person-hours / 1 working day

Initial planning/setup/BL-00/training allowance totals 26 person-hours. Recurring and later-phase effort is additional: multiply the per-occurrence estimates by the approved numbers of reports, meetings, iterations, baselines, audits and releases, avoiding double-counting combined work. Actual time records are compared against these activity-based allowances. The Project Manager approves capacity and re-estimates each phase.

12.2 Schedule and Dependencies

Table 31: Milestones and dependencies

Milestone / timing	Dependency / coordination
SCMP submission — November 10, 2026	Team review and completed document fields; approval status recorded as it actually stands.
Identification submission — November 12, 2026	Inventory/naming document and actual repository/baseline evidence required by instructor.
BL-00 — after reviewed starting artifacts	Cannot precede artifact review and recorded approval.
BL-1 → BL-2	Requirements approval before final design baseline.
BL-3 → BL-4 → BL-5	Implemented-scope checks → acceptance tests/FCA → PCA/release approval.
Weekly / fortnightly	Team meeting weekly; report and minimum CCB scheduling every two weeks; urgent CRs sooner.
Audit and release appointments	Schedule at least two working days before intended approval when feasible, allowing corrective work.
Change control in effect — from repository setup	PR control applies from first capture; formal CR control applies to each item once baselined.
Configuration audits — start and completion	Start at least two working days before the intended baseline approval; complete before the CCB decision.

Later calendar dates are not known; the Project Manager enters them in the controlled project schedule before implementation, updates this plan if dependencies or commitments change and informs the instructor of conflicts.

12.3 Personnel, Tools and Costs

Table 32: Resource allocation

Resource	Allocation / funding assumption / owner
Personnel	Five members supply estimated hours above; no paid labor assumed. Labor cost is 0 project cash expenditure, not zero effort. Any monetary valuation requires an agreed rate.
Development equipment	Member-owned laptops; no new purchase planned. PM confirms availability and backup access.

Resource	Allocation / funding assumption / owner
Repository / automation	Git/GitHub/GitHub Actions; use available allowance only after capability/quota verification. CM/Build Engineer monitor usage.
Hosting	Use an approved available/free academic/demo allocation if sufficient. No paid plan assumed; PM obtains actual quote and approval before expenditure.
Backup storage	Existing separate storage if adequate; CM checks capacity. Any purchase requires an approved amount before commitment.
Software / documentation	Node.js/PostgreSQL, editor, tests, Markdown; Word through an existing authorized license if available, otherwise use an approved compatible editor.
Communication	Existing internet and Telegram; no new dedicated cash allocation assumed.

The initial incremental cash budget assumption is 0 for already available resources, subject to verification; this is not a claim that hosting, software or internet are universally free. Any required paid service is a resource gap: the Project Manager records supplier quote, currency, amount, funding owner and approval before use. Tool versions/configurations and setup instructions are controlled under CI-CFG-01/03/06.

12.4 Training

Train members on branches, PRs, CRs, interfaces, baseline evidence, secrets and recovery; record attendance and a practice PR.

12.5 SCM Tools and Automation

The plan states what is controlled, when and by whom; the tools provide how. Tools are selected to fit the processes in this plan, and a tool change affecting a baseline requires a CR (Section 4.3). Personnel for each activity are given in Section 12.1, equipment and how each resource is obtained in Section 12.3, and training in Section 12.4.

Table 33: SCM functions and tool support

SCM function	Tool	Automated	Remains a team control
Version control and identification	Git / GitHub	History, branches and tags	Baseline tag applied after approval
Change control	GitHub Issues, PRs, labels, board, branch rules	Issue/PR links, review and check enforcement	Impact analysis and CCB decision
Build and verification	GitHub Actions; Jest / Supertest	Install, lint, tests and build	QA review of evidence; audits
Dependency control	Lock files; GitHub dependency alerts where available	Version pinning and advisory notices	Lead triage and acceptance
Status accounting	GitHub records; CI register	Event history	Report preparation; register accuracy
Release and delivery	GitHub Releases; checksums	Package storage and identity	CCB release approval; PCA
Backup	Git bundle; separate storage	Repository export when scripted	Restoration test and record

12.6 Lifecycle Alignment

The project follows a phased, incremental lifecycle: each phase ends at a reviewed baseline, and development proceeds in short iterations. The table aligns project phases with the five project-management phases [R5].

Table 34: SCM alignment with the project lifecycle

Project phase	Management phase	SCM activities	Review point	Baseline
Start-up	Initiation; Planning	Charter, SCMP, CI inventory, repository setup, training.	SCMP and inventory review	BL-00
Requirements	Planning	SRS control, requirement IDs, traceability started.	SRS review	BL-1
Design	Planning	SDD, UI and database design, interface contracts.	Design review	BL-2
Development	Execution	Branch and PR control, automated checks, dependency review.	Phase audit	BL-3
Testing	Execution	Defect tracking, traceability completion, FCA.	FCA	BL-4
Release and demonstration	Execution	Release branch, PCA, delivery and acceptance records.	PCA and CCB approval	BL-5
Handover	Closure	Final archive, handover and retirement.	Archive review	Final archive

The Monitoring and Control phase runs across all project phases through the status reports, measures, surveillance and audits in Sections 6 and 7.

13. Plan Maintenance, Deviations and Risks

13.1 Plan Maintenance

- The Project Manager monitors this plan and reviews it at the start of each phase or iteration and when scope, roles, tools, security requirements or resources change.
- Baseline amendments require a CR, impact analysis, CCB approval, new version/revision-history entry and team notification.
- The Configuration Manager updates controlled copies and supersedes outdated copies.
- QA checks consistency with actual practice, the companion identification document and project schedule.

13.2 Deviations

Record deviations with owner, corrective due date and approval; do not relabel unimplemented controls as completed.

13.3 Risk Register

Table 35: Risk register

Risk	Likelihood	Impact	Response

Risk	Likelihood	Impact	Response
Merge conflicts	High	Medium	Short branches, module ownership and frequent integration.
Lost work	Medium	High	Regular commits, separate backups and restoration checks.
Scope creep	Medium	High	Requirement baseline and CR approval.
Secret exposure	Medium	High	Ignored secret files, review, rotation and incident tracking.
Document/code/interface mismatch	Medium	High	Same-change updates, traceability and contract tests.
Member unavailable	Medium	Medium	Named backup roles and shared records.
Dependency/service change	Medium	High	Supplier monitoring, lock files, change review and recovery plan.
Tool/hosting limits	Medium	Medium	Capability checks, documented manual gates and approved resource alternatives.
Planning gap: later course dates unknown	High	Medium	Phase-relative gates in Section 12.2; Project Manager enters dates once issued.
Documentation control: Word files do not compare well in Git	Medium	Medium	Named owner, revision history, exports only from approved revisions, Markdown where practical.
Coordination: shared roles delay CCB decisions	Medium	Medium	Fortnightly CCB slot, three-vote rule and votes recorded in the CR.
Plan not followed in practice	Medium	High	Surveillance (Section 6.5), SCMP audit (Section 13.4) and corrective actions.

Mitigation is carried out within the Section 12.1 activities, with a proposed allowance of 2 person-hours per two-week period for risk follow-up. The Project Manager reports residual risks to the CCB and, where they affect delivery, to the instructor.

13.4 Control and Audit of the SCMP

- **Controlled item:** this plan is CI-DOC-04 and follows the naming, versioning and change-control rules that apply to any other CI [R4].
- **Distribution and access:** the approved version is held in docs/scm/. The team and instructor have read access, changes enter only through a PR, and the Configuration Manager records each issued version and its recipients.
- **Audit:** each audit before BL-3, BL-4 and BL-5 also checks that this plan is current, that its revision history is complete and that actual practice matches it. Findings follow Section 7.4.

13.5 Continuous Improvement

- **Feedback:** any member may raise a process improvement as a GitHub Issue at any time.
- **Lessons learned:** at each phase boundary the team reviews Section 6.3 measures and surveillance and audit findings, then records what to keep or change in docs/scm/reports/.
- **Updates and core objectives:** accepted improvements are made through Section 13.1 and cannot remove identification, change control, status accounting, auditing or independent review.

14. Deliverables and Approval Readiness

14.1 Deliverables

Table 36: Deliverables and acceptance evidence

Deliverable	Acceptance evidence
Complete SCMP	Project-specific processes, estimates, responsibilities, revision and actual approval record.
CI inventory	Actual paths, owners, versions/statuses and traceable Git references.
Identification/naming document	Consistent rules and full inventory, including CI-DOC-12, CI-CFG-06, CI-REC-03 and CI-REC-04.
Initial baseline	Actual reviewed contents, manifest, approval, exact commit, immutable tag and backup.
GitHub repository	Accessible URL with identified actual artifacts, including available source, docs, tests and configuration.

14.2 Approval Readiness

Before CCB approval, the team completes the following:

- Complete pending fields.
- Approve tailoring and resource assumptions.
- Reconcile the companion inventory.
- Verify repository controls.
- Record actual baseline evidence.

Draft status remains until team approval. Approval records contain only actual approvals and artifacts that actually exist.

Appendix A. IEEE 828-2012 Coverage and Verification Matrix

This matrix maps IEEE 828-2012 clause and Annex D subject areas to sections of this plan. It is a review aid, not a reproduction of the standard. All topics listed for CM planning in clause 6.2.1.2 a) to q) are addressed and none is omitted. Conformance in practice is shown by the evidence recorded as the plan is carried out.

Table 37: IEEE 828-2012 coverage and verification matrix

IEEE reference / area	Plan coverage	Verification evidence to record
Clause 3 — Tailoring	2, 13	Approved tailoring register; exceptions and rationale.
Clause 5 — CM process	3–14	Integrated workflow and linked records.
Clause 6 — CM planning	1–4, 12–13	Approved scope, estimates, schedule, resources and maintenance.
Clause 7 — CM management	3, 6, 12–13	Monitoring reports, allocations and corrective actions.
Clause 8 — Configuration identification	4	Selection criteria, acquired CIs, register and baselines.
Clause 9 — Change control	5	CR decisions, reviewed implementation and verification.
Clause 10 — Status accounting	6	Complete and accessible CI/CR/baseline reports.
Clause 11 — Configuration auditing	7	Audit plans, reports and verified finding closure.
Clause 12 — Interface control	8	Approved contracts and coordinated change/test records.
Clause 13 — Supplier CI control	9	Provenance, acceptance, versions/licenses and update records.
Clause 14 — Release management	10–11	Build, approval, delivery, acceptance and recovery records.
Annex D.1 — Introduction	1	Purpose, scope, lifecycle and references.
Annex D.2 — CI identification criteria	4.1–4.2	Selection/granularity rationale and inventory.
Annex D.3 — Limitations/assumptions	2, 12	Reviewed applicability and resource assumptions.
Annex D.4 — Responsibilities/authorities	3, 5	Named decision and release authority.
Annex D.5 — Project organization	3	Reporting chain, CCB and project interfaces.
Annex D.6 — CM responsibilities	3–13	Owners for activities, tools, verification and backups.
Annex D.7 — Policies/directives/procedures	1.3, 2, 5	Applicable directives and project procedure records.
Annex D.8 — Activities/schedule/resources	4–12	Activity estimates, dependency schedule and approved resources.
Annex D.9 — CMP maintenance	13	Review cadence and controlled revision/communication records.

Appendix B. Reusable Control Forms

B.1 Change Request

Table 38: Change Request form

Field	Entry
CR ID / GitHub Issue; requester/date	To be completed

Field	Entry
Affected CIs, baseline and interface versions	To be completed
Description, justification and priority	To be completed
Current state (Section 5.4)	To be completed
Impact: scope/schedule/effort/cost/security/tests/documents	To be completed
Decision, votes, conditions, date and rationale	To be completed
Assignee, target date, implementation PR/commit	To be completed
Verification evidence, final configuration and closure	To be completed

B.2 Baseline / Release Manifest

Table 39: Baseline / release manifest form

Field	Entry
Baseline/release identifier and status	To be completed from actual evidence
Repository URL; exact approved commit/tag	To be completed from actual evidence
Included CI IDs, paths and versions	To be completed from actual evidence
Build/tool/environment identity	To be completed from actual evidence
Checks, audit findings and accepted deviations	To be completed from actual evidence
CCB approval reference/date	To be completed from actual evidence
Package identity/checksum and delivery location	To be completed from actual evidence
Backup/restoration verification	To be completed from actual evidence
Delivery/acceptance record where applicable	To be completed from actual evidence

B.3 Interface / Supplier Register

- **Interface fields:** ID, nature, owner, affected parties, producer/consumer, contract path/version, compatibility, change reference and tests.
- **Supplier fields:** item/provider, immutable version/origin, license/notices, owner, acceptance criteria/evidence, affected CIs, advisories and update/disposition record.

B.4 Pull Request / Release Checklists

- **PR:** linked issue/CR; naming; affected CIs/interfaces; tests; no secrets or real financial data; document/register updates; independent review and passing checks.
- **Release:** approved scope/deferrals; no critical/high open defects; accepted tests/audits; migrations/compatibility; updated user/setup/license records; approval/tag/manifest; controlled delivery and backup.

Appendix C. Illustrative Requirement Traceability

Examples only: these are not approved requirements, implemented module claims or passed tests. They are replaced with actual SRS, design paths, code and test records.

Table 40: Illustrative requirement traceability

ID	Requirement	Proposed area	Test IDs
FR-01	Register/login with seller, investor and admin roles	Authentication	TC-01, TC-02
FR-02	Upload invoice amount, due date and debtor details	Invoices	TC-03
FR-03	Browse/filter invoices	Marketplace	TC-04
FR-04	Purchase or bid	Trading	TC-05
FR-05	Admin verifies/rejects invoices	Administration	TC-06
FR-06	Record/show payment and settlement status	Settlement	TC-07
FR-07	History and reports	Reporting	TC-08

References

[R1] IEEE Std 828-2012, Standard for Configuration Management in Systems and Software Engineering. IEEE, published March 16, 2012; status Inactive-Reserved since March 30, 2023, and used here because it is the edition assigned for this project. Primary reference. Clause and Annex D requirements were checked against the text of the standard. https://standards.ieee.org/standard/828-2012.html (accessed November 8, 2026).

[R2] NASA Software Engineering Handbook, topic 5.06, SCMP — Software Configuration Management Plan, updated August 1, 2023. Supplementary content and small-project guidance. https://swehb.nasa.gov/spaces/SWEHBVC/pages/50888767/5.06+-+SCMP+-+Software+Configuration+Management+Plan (accessed November 8, 2026).

[R3] Project Part-I assignment instructions, project team roster and companion Configuration Identification Document. Local project artifacts receive their actual approved versions in the CI register.

[R4] Leon, A. (2015). Software Configuration Management Handbook (3rd ed.). Norwood: Artech House. Supplementary guidance on SCMP control and audit.

[R5] Brown, L. (2025, May 20). Understanding the Project Management Life Cycle. Invensis Learning. https://www.invensislearning.com/blog/5-phases-project-management-lifecycle/.

[R6] Federal Democratic Republic of Ethiopia. Personal Data Protection Proclamation No. 1321/2024, in force July 24, 2024. Referenced only as a condition for any future use of real personal data.